Example continued 9/16/2024

Intuitively, high sensitivity of solution to r.h.s. b. Verify theoretically:

$$k_1(A) = \|A\|_1 \|A^{-1}\|_1 = (1999)^2 = 3.996 \times 10^6$$

$$k_\infty(A) = k_1(A)$$

$$k_2(A) = 3.992 \times 10^6$$

(cond(A, norm) in matlab)

So

$$\frac{\|\delta x\|}{\|x\|} \leq 3.99 \times 10^6 \frac{\|\delta b\|}{\|b\|}$$

bound we proved

"Can" be large

usually is

This was highly ill-conditioned system.

Try to quantify the effect of $k(A)$.

Considerations

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta b\|}{\|b\|}, \quad Ax = b$$

- Suppose $\kappa(A) \leq 10^2$. Then if $\frac{\|\delta b\|}{\|b\|} \approx 10^{-4}$, we have

$$\frac{\|\delta x\|}{\|x\|} \leq 10^2 \cdot 10^{-4} = 10^{-2} = \frac{1}{100}\text{th}$$

Sometimes acceptable, other time NO

- Sometimes accuracy of floating point arithmetic is the deciding factor:

Solve $Ax = b + \delta b$, $\frac{\|\delta b\|}{\|b\|} \approx 10^{-7}$

If $\kappa(A) = 10^7$, then $\frac{\|\delta x\|}{\|x\|} = 1$ NOT GOOD

$\kappa(A) = 10^{3,4,5}$ then OK.

<u>Example</u> Famously ill-conditioned matrix

(Matlab has a gallery of ill-conditioned matrices)

$$H_n = [h_{ij}] \in M_n(\mathbb{R})$$

$$h_{ij} = \frac{1}{i+j-1}$$

e.g. $H_4 = \begin{bmatrix} 1 & 1/2 & 1/3 & 1/4 \\ 1/2 & 1/3 & 1/4 & 1/5 \\ 1/4 & 1/5 & 1/6 & 1/7 \end{bmatrix}$

$$\kappa_2(H_4) = 1.6 \times 10^4$$

$$\kappa_2(H_8) = 1.5 \times 10^{10}$$

H_n : <u>Hilbert matrix</u>

Better work harder : how to deal with ill-conditioning.

To understand better $\kappa(A)$:

<u>Definition</u> Given $A \in M_n(\mathbb{C})$

— $\text{maxmag}(A) = \max\limits_{x \neq 0} \dfrac{\|Ax\|}{\|x\|} = \|A\|$

maximum magnification

— $\text{minmag}(A) = \min\limits_{x \neq 0} \dfrac{\|Ax\|}{\|x\|}$

minimum magnification

<u>Proposition</u>

(a) $\text{maxmag}(A) = \dfrac{1}{\text{minmag}(A^{-1})}$

$\text{minmag}(A) = \dfrac{1}{\text{maxmag}(A^{-1})}$

(b) $\kappa(A) = \dfrac{\text{maxmag}(A)}{\text{minmag}(A)}$

Proof

(a) $$\text{maxmag}(A) = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

$$\overset{y := Ax}{=} \max_{y \neq 0} \frac{\|y\|}{\|A^{-1}y\|}$$

$$\overset{!}{=} \frac{1}{\min\limits_{y \neq 0} \frac{\|A^{-1}y\|}{\|y\|}} = \frac{1}{\text{minmag}(A)}$$

Similarly the other conclusion.

(b) $$\frac{\text{maxmag}(A)}{\text{minmag}(A)} \overset{(a)}{=} \text{maxmag}(A) \cdot \text{maxmag}(A^{-1})$$

$$= \|A\| \, \|A^{-1}\| = \kappa(A)$$

■

<u>Corollary</u> ill-conditioning $\Rightarrow$

$$\text{maxmag}(A) \gg \text{minmag}(A)$$

(for nonsingular matrices)

IMPORTANT MESSAGES

- Almost singular matrices ($\det \approx 0$) have minmag ≈ 0 and so they "may" have large condition number (by (b) above)

  So it is tempting to associate ill-conditioning to (almost) singularity.

  NOT TRUE IN GENERAL
  (see the 2x2 example above)

- Ill-conditioning can be caused by "poor scaling" Eg.:

  Consider

$$Ax = \begin{bmatrix} 1 & 0 \\ 0 & \varepsilon \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ \varepsilon \end{bmatrix}$$

$\varepsilon > 0$, "small"

Solution (by hand) is $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

<u>But</u> $\kappa_1(A) = \kappa_2(A) = \kappa_\infty(A) = \dfrac{1}{\varepsilon}$

(ill-cond. A for small ε)

However, take $b + \delta b = \begin{bmatrix} 1 \\ 2\varepsilon \end{bmatrix}$, $\delta b = \begin{bmatrix} 0 \\ \varepsilon \end{bmatrix}$

So $\dfrac{\|\delta b\|_\infty}{\|b\|_\infty} = \dfrac{\varepsilon}{1} = \varepsilon$

Solution to $Ax = b + \delta b$ is $x + \delta x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

WHY?

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(continuing last example)

The radical departure from the exact solution is due to "bad scaling" in $\begin{bmatrix} 1 & 0 \\ 0 & \varepsilon \end{bmatrix}$. How to remedy?

<u>Theorem</u> Let $A \in M_n(\mathbb{C})$ be nonsingular with columns $a_1, a_2, \ldots, a_n \in \mathbb{C}^n$. Then $\forall i, j \in \{1, 2, \ldots, n\}$,

$$\kappa_p(A) \geq \frac{\|a_i\|_p}{\|a_j\|_p}, \quad 1 \leq p \leq \infty$$

<u>Proof</u> Let $a_i = Ae_i$ (e_i = ith column of I)

Then, $\forall i$

$$\max\text{mag}(A) = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} \geq \frac{\|Ae_i\|_p}{\|e_i\|_p} = \|a_i\|_p$$

& $\forall j$

$$\min\text{mag}(A) \leq \cdots \leq \|a_j\|_p$$

So

$$\kappa_p(A) = \frac{\max\text{mag}(A)}{\min\text{mag}(A)} \geq \frac{\|a_i\|_p}{\|a_j\|_p} \quad \blacksquare$$

So if A has columns that differ by several orders magnitude in length, then $\kappa_p(A)$ is large: We refer to this instance as "bad scaling" (of the columns).

<u>Note</u> As $\kappa_\infty(A) = \kappa_1(A^T)$ it is intuitively clear that "bad scaling" (of the rows) is also important.

Typical way to achieve well-scaled rows (or columns) is use scaling by a diagonal matrix

$$Ax = b$$

A has badly scaled rows $r_1^T, r_2^T, \ldots, r_n^T \in \mathbb{C}^n$

Say $\|r_i\|_p \gg \|r_j\|_p$

Define diag. matrix $D = \begin{bmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n \end{bmatrix}$

with $d_i \ll d_j$, $d_i \neq 0$ and solve

$$DAx = Db$$

scales row i by d_i

$i = 1, 2, \ldots, n$

(equivalent system that is better scaled)

We call the transformation $A \to DA$ as a preconditioning technique.

<u>Example</u> (in the previous system)

Instead of $\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & \varepsilon \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 \\ \varepsilon \end{bmatrix}}_{b}$

Solve

$$\begin{bmatrix} 1 & 0 \\ 0 & 1/\varepsilon \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \varepsilon \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1/\varepsilon \end{bmatrix} \begin{bmatrix} 1 \\ \varepsilon \end{bmatrix}$$

$$\Leftrightarrow \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Leftrightarrow \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \checkmark$$

Estimating $\kappa(A)$

In theory $\kappa(A) = \|A\| \, \|A^{-1}\|$.

- $\|A\|$ is "easy" to compute
- $\|A^{-1}\|$ is not easy / expensive to compute with computing A^{-1}.

Can we gauge $\|A^{-1}\|$ without finding A^{-1}?

Trick Say we need $\kappa_1(A)$

Find $\|A\|_1$ ✓

$$\|A^{-1}\|_1 \geq \frac{\|A^{-1}w\|_1}{\|w\|_1}, \quad \forall w \qquad (*)$$

In solving $Ax=b$, take $w=b$ so

$$Ax=w \rightarrow x=A^{-1}w$$

so

$$\frac{\|x\|_1}{\|b\|_1} \leq \|A^{-1}\|_1 \text{ by } (*)$$

i.e.

$$\kappa_1(A) = \|A\|_1 \, \|A^{-1}\|_1$$

$$\geq \|A\|_1 \, \frac{\|A^{-1}w\|_1}{\|w\|_1} \qquad (**)$$

It can be shown that if we choose w in $(**)$ in the direction of max magnification by A^{-1} (see Ex. 2.2.6) then

$$\kappa_1(A) \approx \frac{\|A\|_1 \, \|A^{-1}w\|_1}{\|w\|_1}$$

This of course requires knowing $A^{-1}w$.
But this can be avoided (found) in a simple way:

$$A^{-1}w = x$$

when $Ax = w$

So solve (approximately) using LU to get an estimate for x.

<u>Notes</u>

- Why not just compute $\boxed{\text{minmag}(A)}$ and $\text{maxmag}(A) = \|A\|$ & use

$$\kappa(A) = \frac{\|A\|}{\text{minmag}(A)} \quad ?$$

NO: Because $\text{minmag}(A) = \min_{x \neq 0} \frac{\|Ax\|}{\|x\|}$

no known formula/minimum for this

- Orthogonality helps!

orthogonal (unitary) $A^{-1} = A^T$

Then $\kappa_1(A) = \|A\|_1 \, \|A^{-1}\|_1 = \|A\|_1 \, \|A^T\|_1$

$= \|A\|_1 \, \|A\|_\infty$ ✓

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<u>Note</u> When $A^{-1} = A^T$ in $Ax = b$, and

since $\|Aw\|_2 = \|w\|_2$, $\forall w$

$$\left(\|Aw\|_2^2 = (Aw)^T Aw = w^T A^T A w \right.$$
$$\left. = w^T w = \|w\|_2^2 \right)$$

then $\|x\|_2 = \|b\|_2$. So any small perturbation in $b \rightsquigarrow b + \delta b$ will necessarily lead to a small perturbation in the $\|x\|_2 \sim \|\hat{x}\|_2$

§ 2.3 Perturbing the coefficient matrix

Consider the effect in solving $Ax = b$

when $A \rightsquigarrow A + \delta A$

and $\dfrac{\|\delta A\|}{\|A\|}$ is "small"

Denote $(A + \delta A)\hat{x} = b$.

Want conditions so that x & $\hat{x}$ are close.

<u>Theorem</u> If $A \in M_n(\mathbb{C})$ is nonsingular

and $\dfrac{\|\delta A\|}{\|A\|} < \dfrac{1}{\kappa(A)}$, then

$A + \delta A$ is nonsingular.

Proof

Prove the contrapositive.

Suppose $A+\delta A$ is singular.

So $\exists\, y \neq 0 \;:\; (A+\delta A)y = 0$

$$\Rightarrow \quad Ay = -\delta A y$$

$$\Rightarrow \quad y = -A^{-1}\delta A y$$

$$\Rightarrow \quad \|y\| = \|A^{-1}\delta A y\|$$

$$\leq \|A^{-1}\|\,\|\delta A\|\,\|y\| \qquad (\|y\|>0)$$

$$\Rightarrow \quad 1 \leq \|A^{-1}\|\,\|\delta A\|$$

$$\Rightarrow \quad \frac{1}{\kappa(A)} = \frac{1}{\|A\|\,\|A^{-1}\|} \leq \frac{\|\delta A\|}{\|A\|} \qquad \blacksquare$$

The above inequality can be difficult to satisfy in $\kappa(A)$ is large.

Notes

(1) Interpret the above as providing a "distance to singularity" via $\kappa(A)$

A nonsingular & $A+\delta A$ is singular

$$\Longrightarrow \quad \frac{1}{\kappa(A)} \leq \frac{\|\delta A\|}{\|A\|}$$

Equality holds for $\kappa_2(A)$, $\|A\|_2$ (Cor 4.3.22)

(2) Sufficient condition in this theorem is difficult to satisfy if $\kappa(A)$ is large.

Back to the effect of δA in solving $Ax=b$:

$$(A+\delta A)\hat{x} = b$$
$$\delta x := \hat{x} - x$$

How do know that

$$\frac{\|\delta x\|}{\|x\|} \text{ is small?}$$

Study $\frac{\|\delta x\|}{\|x\|}$ first.

<u>Theorem</u> Let $A \in M_n(\mathbb{C})$ is nonsingular, $b \in \mathbb{C}^n$, $b \neq 0$. Consider the systems

$$Ax = b \quad \& \quad (A+\delta A)\hat{x} = b$$

when $\delta A \in M_n(\mathbb{C})$. Then

$$\boxed{\frac{\|\delta x\|}{\|\hat{x}\|} \leq \kappa(A) \frac{\|\delta A\|}{\|A\|},}$$

where $\delta x = \hat{x} - x$, $\kappa(A) = \|A\| \, \|A^{-1}\|$.

<u>Proof</u>

$$\left.\begin{aligned} Ax &= b \\ (A+\delta A)\hat{x} &= b \\ \hat{x} &= x + \delta x \end{aligned}\right\} \Rightarrow \delta x = -A^{-1}\delta A\, \hat{x}$$

So $\|\delta x\| \leq \|A^{-1}\| \, \|\delta A\| \, \|\hat{x}\|$

$$\Rightarrow \frac{\|\delta x\|}{\|\hat{x}\|} \leq \|A^{-1}\| \|\delta A\| = \kappa(A) \frac{\|\delta A\|}{\|A\|} \quad \blacksquare$$

$(\hat{x} \neq 0$, otherwise $b = 0)$

Theorem (main)

If $A \in M_n(\mathbb{C})$ is nonsingular, $\frac{\|\delta A\|}{\|A\|} < \frac{1}{\kappa(A)}$, $b \neq 0$, $Ax = b$,

$(A + \delta A)\hat{x} = b$, then

$$\boxed{\frac{\|\delta x\|}{\|x\|} \leq \frac{\kappa(A)\,\frac{\|\delta A\|}{\|A\|}}{1 - \kappa(A)\,\frac{\|\delta A\|}{\|A\|}}}$$

Proof As in previous theorem

$$\|\delta x\| \leq \|A^{-1}\|\,\|\delta A\|\,\|\hat{x}\|$$

$$(\hat{x} = \delta x + x)$$

$$\overset{\text{triangle}}{\leq} \|A^{-1}\|\,\|\delta A\|\,(\|x\| + \|\delta x\|)$$

$$\Rightarrow \underbrace{\left(1 - \kappa(A)\,\frac{\|\delta A\|}{\|A\|}\right)}_{\neq 0}\|\delta x\| \leq \kappa(A)\,\frac{\|\delta A\|}{\|A\|}\,\|x\|$$

∎

Combine perturbations

$$b \rightsquigarrow b+\delta b$$

$$A \rightsquigarrow A+\delta A$$

$$\boxed{(A+\delta A)\hat{x} = b+\delta b}$$

Theorem Let $A \in M_n(\mathbb{C})$ be nonsingular,

$$\frac{\|\delta A\|}{\|A\|} < \frac{1}{\kappa(A)}, \quad b \neq 0, \quad Ax = b,$$

$$A(\underbrace{x+\delta x}_{\hat{x}}) = b+\delta b. \quad \text{Then}$$

$$\frac{\|\delta x\|}{\|x\|} \leq \frac{\kappa(A)\left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|b\|}\right)}{1-\kappa(A)\left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|b\|}\right)}$$

Proof See ex. 2.3.12 / 2.3.13